

Quantised failure: trained reflection transports collapse onto exact group quotients

Technical report — draft v1

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Abstract

A recurrent layer whose state is moved by products of Householder reflections can be trained on the word problem of a finite group: read a sequence of group elements, report the product. When such a layer succeeds it is exact at any length, because it has learned a homomorphism. When it fails, it does not fail arbitrarily. Its accuracy lands on $1/|N|$ for a normal subgroup N — it has learned the quotient G/N , gets the coset right, and guesses inside it. We give the proven ceiling that makes the phenomenon measurable, the rung structure across several groups, a control at matched group order that separates the lattice from the order, a table-free diagnostic based on characters, and the tasks where a quotient is sufficient rather than a failure. We also report the same invariant appearing in an unrelated role: the normalised trace that identifies which quotient a transport learned is the quantity that governs how much information the same transport can bind and store.

1 The phenomenon

Train the transport of Section 2 on the word problem of S_4 and it either reaches 1.000 and stays there as the evaluation length grows to $8\times$ the training length, or it settles on one of a small set of values: 0.250, 0.083, 0.042. Those are $1/4$, $1/12$, $1/24$, and the normal subgroups of S_4 are 1, V_4 (order 4), A_4 (order 12) and S_4 itself. A model that has learned G/N knows which coset the answer lies in and cannot do better than guessing inside it, so its accuracy is exactly $1/|N|$.

The claim of this paper is that this is what failure looks like for this architecture, that it can be verified rather than inferred from a coincidence of four numbers, and that the same algebra predicts a second, apparently unrelated limitation of the same layer.

What makes it checkable. Three things, in increasing strength.

1. **Flatness.** A homomorphism is length-independent. Every claim below is scored at a length far beyond training, and runs whose accuracy drifts with length are excluded rather than scored: they are not homomorphisms and the law says nothing about them.
2. **Coset consistency.** If the model learned G/N , its *errors* must respect cosets: predicted and true element in the same coset of N , essentially always. A model that is merely bad has no reason to respect any coset.
3. **A proven ceiling.** For a commuting transport the product is order-independent, so the output can only be a function of the multiset of inputs; the best such function is computable by enumeration and is a hard bound.

2 Setup

The setup is deliberately not the language model of paper A. There is no input projection and no gate: each group element g owns a learned table of n_h reflection vectors $\mathbf{u}_{g,1..n_h} \in \mathbb{R}^d$ and coefficients $\beta_{g,i}$, and the

transport it induces is their product,

$$T_g = \prod_{i=1}^{n_h} (I - \beta_{g,i} \hat{\mathbf{u}}_{g,i} \hat{\mathbf{u}}_{g,i}^\top), \quad \hat{\mathbf{u}} = \mathbf{u} / \|\mathbf{u}\|,$$

initialised at $\beta = 2$, i.e. at pure reflections. The state starts at a learned $\mathbf{h}_0$, each token applies its own T_g , and a linear readout maps the state to the $|G|$ classes. Giving every element its own transport removes the question of whether a shared projection can separate the elements and leaves only the question this paper asks: can a *product of reflections* realise the group?

Training reads a word $g_1 \dots g_L$ and is supervised on the running product at every prefix, with an explicit group-law penalty added to the cross-entropy:

$$\mathcal{L} = \text{CE}(\text{readout}(\mathbf{h}_t), g_1 \dots g_t) + \lambda \mathbb{E}_{\mathbf{h},a,b} \|T_b T_a \mathbf{h} - T_{ab} \mathbf{h}\| / \sqrt{d}, \quad \lambda = 1,$$

the expectation taken over 64 random triples per step. The penalty asks directly for a homomorphism rather than hoping one emerges, which is what makes a failure interpretable: a model that ends with a small penalty and sub-unit accuracy has learned *some* homomorphism, and the question is which.

Unless stated otherwise: $n_h = 4$, $d = 4$ for S_4 and Q_8 and $d = 5$ for A_5 , AdamW at 2×10^{-3} with weight decay 0.01 and gradients clipped at 1.0, batch 256, 2000 steps, trained at length $L = 12$ and evaluated at $L = 12$ and $L = 96$ on 1024 held-out words each. A run counts as *flat*, and so as a candidate homomorphism, when the two accuracies differ by less than 0.05.

3 The abelian ceiling

Before asking which quotient a transport learns, it is worth fixing a bound that no amount of training can move, because it converts one architectural property — whether the transports commute — into a number.

Proposition 1 (multiset bound) *If the transports commute, the accuracy on the length- L word problem of G cannot exceed*

$$C(G, L) = \sum_M \Pr[M] \max_{g \in G} \Pr[g_1 \dots g_L = g \mid M],$$

the sum running over multisets M of L elements.

Proof. If $T_a T_b = T_b T_a$ for all a, b , the state after reading a word depends on the word only through how many times each element occurs, so the prediction is a function of M . The best such function returns, for each M , the most probable product, which is the stated sum. $\square$

The bound is computable by enumeration, and at $L = 8$ we evaluate it exactly over all 1287 multisets of six elements: $C(\mathbb{Z}_6, 8) = 1.0000$ and $C(S_3, 8) = 0.3850$. The pair is matched in order, so a transport that clears 0.385 on S_3 has done something order-sensitive, and nothing about the size of the group can explain it away.

Table 1: Matched-order test at $L = 8$, three seeds. The ceiling is proven, not fitted. Commuting transports sit on it; the reflection transport passes it on S_3 .

transport	commuting?	$\mathbb{Z}_6$	S_3
diagonal rotation	yes	1.000 ± 0.000	0.388 ± 0.006
single Givens	yes	1.000 ± 0.000	0.389 ± 0.003
Householder product ($n_h = 4$)	no	1.000	0.778
butterfly	no	1.000 ± 0.000	0.779 ± 0.380
transformer	—	0.166 ± 0.004	0.167 ± 0.003
<i>proven ceiling for any commuting transport</i>		1.000	0.385

Two honest qualifications. The ± 0.380 on the butterfly row is a spread across seeds wide enough that some seeds solve the task and others fail below the ceiling; the mean exceeding 0.385 says the ceiling *can* be

passed, not that it reliably is. And clearing the ceiling on S_3 does not imply clearing it everywhere: on the same ladder the reflection transport reaches 0.374 on Q_8 against a ceiling of 0.544, 0.085 on A_4 against 0.286, and chance on S_4 and A_5 . Being non-commuting removes a ceiling; it does not supply an optimiser that finds what lies above it. The rest of this paper is about where those runs land instead.

4 The lattice law

A transport that ends training with a small group-law penalty is a homomorphism $\rho : G \rightarrow O(d)$ composed with a readout. Its kernel $N = \ker \rho$ is a normal subgroup, elements of the same coset of N are indistinguishable to it, and its accuracy is therefore $1/|N|$. The set of achievable accuracies is thus the set $\{1/|N| : N \trianglelefteq G\}$ — the *rungs* — which is a property of the group alone and is invisible to the model. We enumerate the normal subgroups of each group directly (unions of conjugacy classes closed under the product) rather than quoting them: S_4 gives orders 1, 4, 12, 24; Q_8 , being Hamiltonian, gives 1, 2, 4, 4, 4, 8; A_5 , being simple, gives only 1, 60.

Table 2: Four seeds per group. Every run is flat, so every run is a candidate homomorphism, and every one lands on a rung of its own group. Worst deviation across all twelve runs: 0.021.

group (rungs)	seed	$L = 12$	$L = 96$	nearest rung (dev.)
Q_8 1.000 0.500 0.250 0.125	0	0.247	0.241	0.250 (0.009)
	1	0.498	0.521	0.500 (0.021)
	2	1.000	1.000	1.000 (0.000)
	3	0.264	0.243	0.250 (0.007)
S_4 1.000 0.250 0.083 0.042	0	1.000	1.000	1.000 (0.000)
	1	0.081	0.100	0.083 (0.017)
	2	0.264	0.251	0.250 (0.001)
	3	0.257	0.259	0.250 (0.009)
A_5 1.000 0.017	0	0.015	0.020	0.017 (0.003)
	1	0.009	0.021	0.017 (0.004)
	2	0.018	0.014	0.017 (0.003)
	3	0.015	0.016	0.017 (0.001)

Which rungs are actually informative. The bottom rung $1/|G|$ coincides with chance, so a run landing there is consistent with the law but tells us nothing: it is the trivial quotient and the absence of learning at once. This applies to all four A_5 runs, at 0.017. The registered falsifier for A_5 — a flat accuracy strictly between 0.017 and 1.000, which a simple group forbids — was not triggered, but neither did any A_5 run solve the task, so the group contributes a consistency check rather than a positive result. The load is carried by the intermediate rungs that are neither chance nor perfect: Q_8 at 0.500 and 0.250, and S_4 at 0.250 and 0.083.

Coset consistency. Accuracy alone could be a numerical coincidence, so we test the mechanism on the errors. If a model learned G/N , its predictions must land in the correct coset of N essentially always, and the kernel is identified as the *smallest* N whose coset score reaches 1.000 — read off the errors, with no reference to the accuracy. The accuracy predicted by that independently identified N then matches the measured one on all five S_4 runs tested:

Table 3: The subgroup identified by the errors predicts the accuracy. Coset scores are the fraction of predictions landing in the correct coset.

raw acc.	1	V_4	A_4	S_4	identified N	$1/ N $
1.000	1.000	1.000	1.000	1.000	1 (faithful)	1.000
1.000	1.000	1.000	1.000	1.000	1 (faithful)	1.000
0.230	0.230	0.999	0.999	1.000	V_4	0.250
0.081	0.081	0.329	1.000	1.000	A_4	0.083
0.042	0.042	0.165	0.472	1.000	S_4	0.042

A measurement we withdraw. Our earlier runs also reported a per-run kernel size, obtained by clustering the transports $M[g] = \mathbb{E}_{\mathbf{h}} T_g \mathbf{h}$ and setting $|\ker| = |G|/(\#\text{clusters})$. That estimator is invalid: T_g is linear and $\mathbf{h}$ is zero-mean, so $M[g]$ is a Monte-Carlo zero for every g and the clustering runs on sampling noise. Measured, $\|M[g]\|$ is 5.7% of the typical $\|T_g \mathbf{h}\|$, below the clustering threshold; applied to an *untrained* model whose 24 transports are independent random draws and whose true kernel is trivial, it reports $|\ker| = 8$. It returned orders that are not normal subgroup orders at all — 8 for S_4 , 30 for A_5 — in 4 of 12 runs, which is how the fault was found. Nothing in this paper rests on that column; the kernel is identified by coset consistency instead.

5 The same-order control

The cross-group evidence is confounded by group order, since a 24-element problem and an 8-element problem differ in chance level as well as in lattice. The control removes it: S_4 and $\text{SL}(2, 3)$ both have order 24, and their rungs are disjoint apart from the endpoints.

$$S_4 : 1.000, 0.250, 0.083, 0.042 \quad \text{SL}(2, 3) : 1.000, 0.500, 0.125, 0.042$$

Across four registered runs (2400 steps, 1200 steps, an S_4 -only run at 1800 steps, and a run at width 3), 28 failures were flat enough to score. Every one landed on a rung of its own group, 13 of them on a rung the other group does not possess, and none within 0.03 of the other group’s exclusive rungs. One run missed its registered tolerance — $\text{SL}(2, 3)$ at 0.469 against a predicted 0.500, with coset consistency 1.000 — so the registered verdict is *falsified*, and we report it as such; in binomial standard errors that miss is $z = -2.00$, the only $|z| > 1.96$ among the 28, which says the tolerance was set too tight rather than that the law failed. The runs, their predictions and their verdicts are in the repository.

6 Characters as a table-free diagnostic

Identifying the kernel by coset consistency needs the group table. There is a diagnostic that does not. For a representation ρ with character $\chi(g) = \text{tr } \rho(g)$, the norm

$$\langle \chi, \chi \rangle = \frac{1}{|G|} \sum_{g \in G} \chi(g)^2 = \sum_i m_i^2$$

counts irrep multiplicities, and needs only traces of the learned transports — no labels, no Cayley table, no character table. Enumerating every way to build a 4-dimensional representation of S_4 and intersecting the kernels of its parts gives what each value implies:

kernel	quotient realised	attainable $\langle \chi, \chi \rangle$ at $d = 4$
1	S_4 (faithful)	2
V_4	S_3	3, 4, 5
A_4	$\mathbb{Z}_2$	8, 10, 16
S_4	trivial	16

At $d = 4$ the value 2 is attainable *only* by a faithful representation, so it is a sufficient certificate of faithfulness. That statement is conditional on full rank, and the condition is not decorative: a trained transport may let a dimension die, and at $d = 3$ the value 2 means kernel V_4 , at $d = 2$ kernel A_4 . We observe exactly this — runs that solve the task with $\langle \chi, \chi \rangle \approx 1.12$ and a single 3-dimensional irrep carrying the whole state. The diagnostic must therefore be read together with the effective rank, not on its own.

A second condition matters more. $\langle \chi, \chi \rangle$ is a character norm only if the transport *is* a representation. In our runs the solved models carry a group-law violation of ≈ 0.008 and measure $\langle \chi, \chi \rangle = 1.99\text{--}2.01$, agreeing with the table; the failed runs carry violations of $0.10\text{--}0.29$, are not homomorphisms at all, and their measured values (4.4–5.5) are not character norms of anything and should not be read as quotient labels. The predicted quotient values are so far confirmed in one clean instance rather than as a set: a model that solves the V_4 task exactly (Section 8) decomposes as two copies of the trivial irrep plus the 2-dimensional one, giving $\langle \chi, \chi \rangle = 2^2 + 1^2 = 5$ with kernel V_4 , as the table requires.

Finally, the diagnostic does not work as a training signal. An arm that adds a penalty driving the mean squared trace toward 2.0 — no table, no labels, nothing but traces — solved 2 of 6 seeds against 3 of 6 for the unregularised baseline. It certifies a representation; it does not produce one.

7 The same invariant governs binding

The diagnostic above is a character: $\chi(g) = \text{tr}(g)$, and $\langle \chi, \chi \rangle$ identifies the learned quotient without the group table. The same quantity controls a capability that has nothing to do with group word problems.

For an orthogonal g acting on the sphere, the expected overlap of a random unit vector with its image is the normalised trace,

$$\mathbb{E}[\mathbf{v}^\top g \mathbf{v}] = \text{tr}(g)/d,$$

which is exactly how much an operation *hides* a vector — the property a binding operation needs if a value is to be stored against a key and recovered later. A single Householder reflection has $\text{tr} = d - 2$, the least-hiding non-trivial element of $O(d)$, and a product of k reflections gives overlap $\approx e^{-2k/d}$. Measured at head dimension 64, with weights set by hand and no training, capacity follows the trace exactly: four associations are recalled at 0.453 with $k = 2$ and at 1.000 with $k = 32 \approx d/2$.

So the same transport is a tracker at small k and an associative store at $k \approx d/2$, and the character is what separates the regimes. A layer that composes group elements perfectly and cannot retain one fact among four is not exhibiting two unrelated limitations; it is at one end of a single curve.

8 What a quotient is good for

Calling a quotient a failure assumes the task needed the whole group. Change the target to the coset of N and the same collapse becomes the correct answer: the V_4 task asks for the V_4 -coset of the product (6 classes, chance 0.167) and needs only $S_4/V_4 \cong S_3$; the A_4 task asks for the A_4 -coset (2 classes, chance 0.500) and needs only $S_4/A_4 \cong \mathbb{Z}_2$.

Solve rates rise as the required group shrinks — 2 of 4 seeds on the full task, 2 of 4 on V_4 , 3 of 4 on A_4 — but the sharper result is where the *failures* land. If the law of Section 4 is really about the group being learned, then on a quotient task it should apply to the quotient, and the rungs should be those of S_3 and $\mathbb{Z}_2$ rather than those of S_4 . The normal subgroups of S_3 have orders 1, 3, 6, giving rungs 1.000, 0.333, 0.167; those of $\mathbb{Z}_2$ give 1.000 and 0.500. Neither 0.333 nor 0.500 is a rung of S_4 .

Table 4: The lattice law one level down. All eight runs land on a rung of the *quotient* they are asked for; worst deviation 0.013.

task	group needed	$L = 96$ accuracy by seed			
V_4 -coset	S_3 (rungs 1.000 0.333 0.167)	1.000	1.000	0.320	0.336
A_4 -coset	$\mathbb{Z}_2$ (rungs 1.000 0.500)	1.000	0.495	1.000	1.000

The failures sit at 0.320 and 0.336 against S_3 's rung of $1/3$, and at 0.495 against $\mathbb{Z}_2$'s rung of $1/2$. A registered prediction accompanies this: a quotient task does not require a faithful representation, so the amplitude on S_4 's 3-dimensional irreps should stay near zero even when the task is solved perfectly. It does — at most 0.042 across all eight quotient runs, against ≈ 1.00 on the solved full-task runs. Capacity tracks the task, not the ambient group.

9 Limitations

Small groups, small dimensions, single-layer transports, four seeds per cell, and a training objective that includes an explicit group-law penalty — the law describes where this architecture lands when it fails, not a claim about trained networks in general. The same-order control is registered as falsified on one run of 28. The Frobenius–Schur account of *which* rung is reached, present in earlier notes, is not claimed here: $SL(2, 3)$ solved the task outright at width 6, which that account does not predict.

Three weaknesses are specific. The bottom rung is chance, so the four A_5 runs confirm the law without demonstrating it, and the positive evidence rests on the intermediate rungs of Q_8 and S_4 and on coset consistency. Clearing the abelian ceiling is shown on S_3 only; on larger groups the same transport falls below what a commuting transport achieves, so Section 3 bounds what the architecture can express and says nothing about what training reliably finds. And the character diagnostic is doubly conditional — on the transport being a homomorphism, and on the representation being full rank — with the predicted quotient values so far confirmed in a single clean instance rather than across the table.

Every derived quantity above — the ceiling, the rung sets, the attainable character norms — is recomputed from the group definitions by `evidence/verify_paperB.py`, which also re-reads the measured tables from the result files rather than trusting a transcription.